

7 Parkinson's disease is a nervous system disorder that affects movement.

(a) In people with Parkinson's disease, some neurones in the brain die.

Many of the symptoms of Parkinson's disease are due to a loss of neurones that produce a neurotransmitter called dopamine.

(i) Describe how a neurotransmitter transmits a nerve impulse across a synapse.

(4)

(ii) One treatment for Parkinson's disease is a drug called L-DOPA.

Describe how L-DOPA reduces some of the symptoms of Parkinson's disease.

(3)

(b) Scientists can induce pluripotent stem cells to produce transcription factors involved in the synthesis of dopamine.

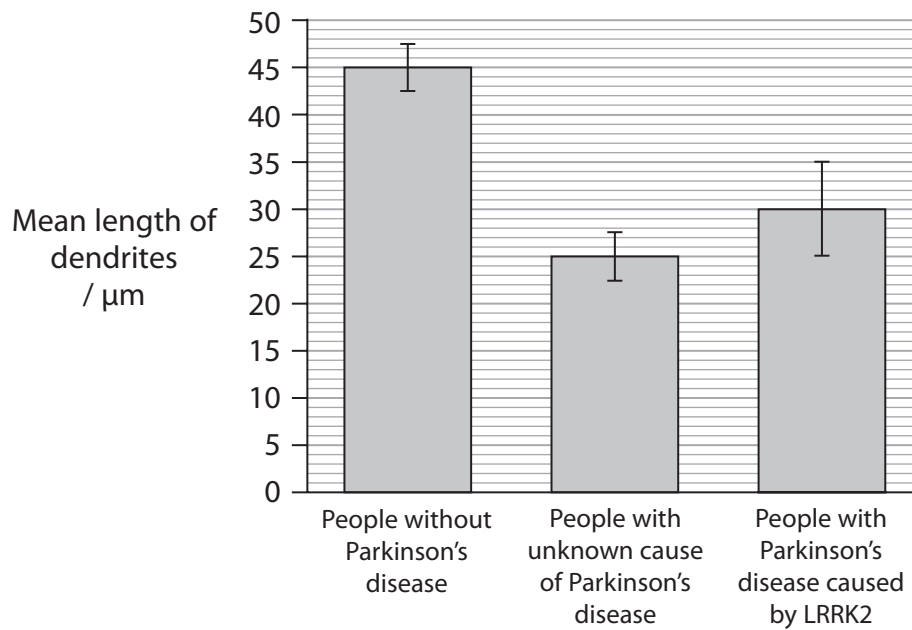
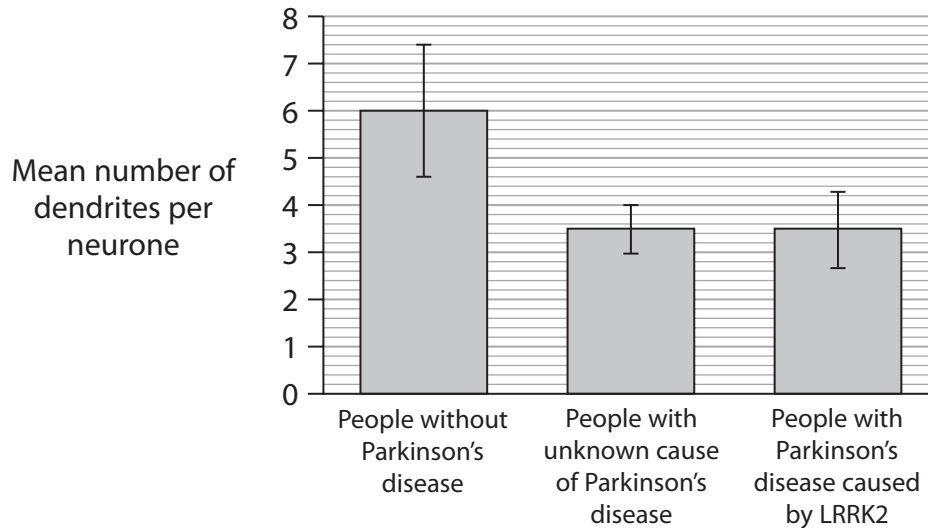
(i) Explain how transcription factors are involved in the synthesis of dopamine.

(2)

(ii) Some people with Parkinson's disease have a DNA mutation called LRRK2.

Scientists studied the effect of this mutation on the dendrites of dopamine neurones produced from the stem cells of three groups of people.

The graphs show some of the results of their study.



Comment on these results.

(3)

(Total for Question 7 = 12 marks)